

Anomaly Detection in Network Traffic

A Machine Learning Approach – Deliverable 3

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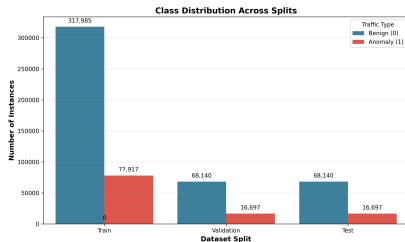
Context and Dataset (CIC-IDS2017)

- **Cybersecurity Challenge:** Identifying malicious traffic dynamically.
- **Dataset Selection:** Canadian Institute for Cybersecurity (CIC-IDS2017).
 - ▶ Realistic normal traffic mixed with modern multi-vector attacks (DDoS, Brute Force, Web Attacks, etc.).
- **Computational Hurdles:**
 - ▶ Over **2.8 million instances** (> 1 GB raw CSVs) triggering OOM errors on local machines.
 - ▶ Severe class imbalance: **80.3% benign** vs. **19.7% anomalous**.

Exploratory Data Analysis (EDA)

Key Insights:

- Highly skewed attributes (Flow Duration: μ s to ≥ 100 s).
- Strong correlations detected between packet volumes and rate features.
- Identified 8 static features (zero-variance) that were pruned.



Native 80/20 Class Distribution

Data Preprocessing Pipeline

- ➊ **Data Cleaning:** Handled column header whitespaces, dropped NaN and Infinite values ($< 0.1\%$ impact).
- ➋ **Feature Selection:** Dropped zero-variance variables (e.g., Bwd PSH Flags) to reduce computational noise.
- ➌ **Stratified Sampling:** Extracted a representative 20% sample ($\approx 565,000$ rows) maintaining the 80/20 class split.
- ➍ **Split & Scale:** 70/15/15 train/val/test split. Normalization using *StandardScaler* fitted strictly on training data to prevent **data leakage**.

Evaluation Strategy

- **Imbalance Impact:** Traditional accuracy is misleading (always predicting 'normal' achieves 80.3% accuracy).
- **Selected Metrics:** F1-Score, Recall (critical to catch all attacks), Precision, and AUC-ROC.
- **Multi-Perspective Framework:**
 - ① **Original Distribution:** Realistic test environment.
 - ② **Balanced Subset (50-50):** Exposes pure algorithm discrimination.
 - ③ **Per-Class Breakdown:** Recall analysis across different attack vectors.

State of the Art in Network Anomaly Detection

Table: Summary of literature models and findings.

Method	Type	Data Benchmark	AUC
Isolation Forest (2008)	Tree Ensemble	KDD'99	0.96
LOF (2000)	Density Ratio	Synthetic	0.89
One-Class SVM (2001)	Kernel Boundary	USPS	0.87
Autoencoder (2014)	Deep learning	MSDS	0.93
Kitsune (2018)	Ensemble AE	Real Traffic	0.99
DAGMM (2018)	Deep Hybrid	KDD'99	0.98

Major Conclusions from Review:

- Tree ensembles and neural networks consistently outperform density methods in dimensions > 20 .
- Hyperparameter tuning is rarely reported systematically in cybersecurity papers.

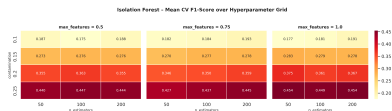
Classical Algorithms Evaluated

- **Isolation Forest:** Builds an ensemble of isolation trees. Anomalies are isolated closer to the root of the trees.
- **One-Class SVM:** Constructs a maximum-margin hyperplane in a high-dimensional kernel space to separate normal points.
- **Local Outlier Factor (LOF):** Compares the local reachability density of a point against its k -nearest neighbors.

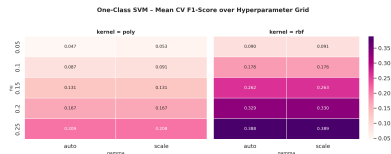
Hyperparameter tuning strategy:

- **3-Fold Stratified Cross-Validation (CV)** conducted on the training split to select optimal parameters.
- Fixed held-out test set preserved strictly for final comparisons.

Classical Models Tuning Heatmaps



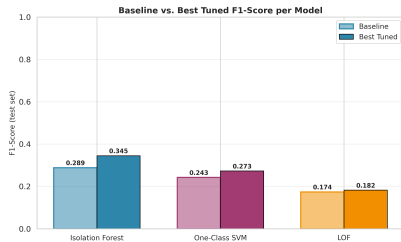
Isolation Forest CV F1-Score



One-Class SVM (RBF vs. Poly) CV F1

- **Isolation Forest:** Contamination is the most sensitive parameter.
- **One-Class SVM:** RBF kernel consistently beats Polynomial.

Baseline vs. Tuned Performance



Test F1 Comparison (Baseline vs. Best)

Results of Tuning:

- **Isolation Forest:** Significant boost from tuning (+5.6% F1 score).
- **One-Class SVM:** Moderate improvement (+3.0% F1).
- **LOF:** Negligible change. Confirming that LOF's density ratios break down in high-dimensional (70D) spaces.

Deep Learning Architectures (Autoencoders)

- **Primary Autoencoder (Normal-Trained):**

- ▶ Trained exclusively on benign network traffic ($y = 0$).
- ▶ Reconstruction MSE loss: $\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum \|\mathbf{x} - \hat{\mathbf{x}}\|^2$.
- ▶ Anomaly Score: $s_{\text{primary}}(\mathbf{x}) = \|\mathbf{x} - \hat{\mathbf{x}}\|^2$.

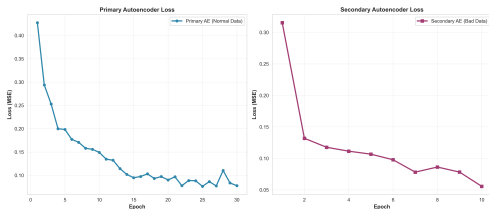
- **Secondary Autoencoder (Anomaly-Trained):**

- ▶ Trained exclusively on malicious traffic ($y = 1$).
- ▶ Learns statistical similarities of attacks.
- ▶ Anomaly Score: $s_{\text{secondary}}(\mathbf{x}) = -\|\mathbf{x} - \hat{\mathbf{x}}_2\|^2$.

- **Ensemble Autoencoder:**

- ▶ Logical OR-gate decision framework: $\hat{y}_{\text{ensemble}} = \hat{y}_{\text{primary}} \vee \hat{y}_{\text{secondary}}$.
- ▶ Maximizes detection rates, minimizing dangerous false negatives.

Autoencoders Training Convergence



Training Loss curves for both network branches

Convergence Analysis:

- Primary AE converged smoothly within 30 epochs (final MSE: 0.0615).
- Secondary AE converged quickly (10 epochs) due to structured attack patterns.
- No overfitting observed on validation set.

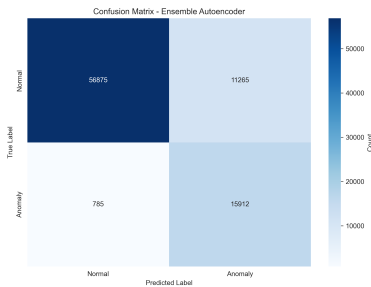
Final Metric Comparison (Test Set)

Table: Model results under original and balanced test splits.

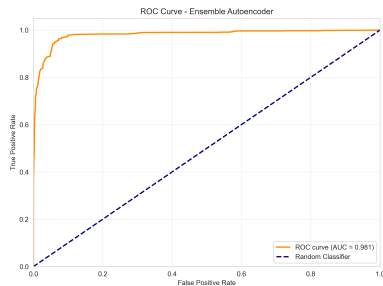
Model	Original (Imbalanced)			Balanced (50-50)		
	Precision	Recall	F1	Precision	Recall	F1
Isolation Forest	0.4522	0.4625	0.4573	0.7770	0.4625	0.5798
One-Class SVM	0.3844	0.3943	0.3893	0.7159	0.3943	0.5085
LOF	0.1327	0.1372	0.1349	0.3902	0.1372	0.2030
Autoencoder	0.5545	0.5635	0.5589	0.8366	0.5635	0.6734
Second AE	0.7781	0.7907	0.7844	0.9375	0.7907	0.8579
Ensemble AE	0.5855	0.9530	0.7253	0.8505	0.9530	0.8989

- **Deep Learning Dominance:** PyTorch Autoencoders far exceed classical.
- **Ensemble AE:** Best recall (**95.3%**) and AUC-ROC (**0.981**), crucial for maximum security.
- **Second AE:** Best precision (**77.81%** original) and F1 (**0.7844**), minimizing false alarms.

Ensemble Diagnostics



Confusion Matrix



ROC Curve (AUC = 0.981)

- Out of 16,697 actual anomalies in the test set, only 785 are missed.
- The 58.55% precision is due to the logical OR aggregating false positives, which is an acceptable compromise for security operations.

Overfitting and Underfitting Analysis

- **Underfitting:**

- ▶ *Local Outlier Factor (LOF)*: Underfits severely in 70D because distance concentration hides outlier patterns.
- ▶ *Baselines*: Isolation Forest and OC-SVM default parameters fail to capture multi-vector attack profiles.

- **Overfitting:**

- ▶ *One-Class SVM*: Highly prone to overfitting if γ is too high, leading to a tight envelope and massive false alarms.
- ▶ *Autoencoders*: Prevented from overfitting (identity mapping) by a compact bottleneck architecture (70 $\rightarrow$ 8).

- **Regularization**: Managed by monitoring train/val loss curves.

Critical Reflection and Responsible AI

- **Theory vs. Practice:** In high dimensions, density concepts (like LOF) degrade. Modeling must align with data geometry.
- **Ethical Dimensions:**
 - ▶ False positives can block legitimate users, impacting business.
 - ▶ Models are trained on university network traffic from 2017; generalizability to modern, zero-day corporate networks requires validation.
- **AI Tools Disclosure:** Used Gemini Pro for debugging PyTorch tensors and Scite.ai/Elicit.com for literature reference verification.

Conclusions and Future Work

- **Key Takeaways:**

- ① Unsupervised Deep Learning is highly successful for anomaly-based IDS.
- ② Ensemble AE provides maximal security monitoring (95.3% recall).
- ③ Secondary AE minimizes false alarms, suitable for automated mitigation.

- **Future Work:**

- ▶ Integrating Deep Autoencoding Gaussian Mixture Models (DAGMM).
- ▶ Adding Attention layers inside the encoder/decoder stacks.
- ▶ Implementing weighted scoring logic instead of strict binary OR.

Thank you. Questions?